DADS 5250 · MODULE 7 HACKATHON · TOPIC BRIEF

Dynamic Agent Staffing Optimization

Staffing support agents for new Google advertisers, month by month, under uncertain demand. A problem contributed by Google.

The problem	Decide how many support agents each country needs in each month of 2025.
Core difficulty	Uncertain daily sign-ups, a 10-advertiser capacity per agent, a 60-day waiting pool, and the cost and delay of hiring and firing.
Data	Advertiser sign-ups per country for 2023 and 2024, plus salaries, eligibility thresholds, and January 2025 staffing. Provided by the teaching team.
GenAI angle	A planning assistant, a scenario explorer, or an automated staffing memo built on top of your model (see Section 5).
Due	Sunday, October 25, 11:59 PM ET

1 Background

Millions of businesses advertise on Google, with thousands of new advertisers joining daily worldwide. To ensure excellent support, Google assigns dedicated agents to assist eligible advertisers for 60 days shortly after they sign up.

Agent staffing is managed locally in each country, with agents serving advertisers in their respective regions. Each agent can handle up to 10 advertisers at any time, but the unpredictable daily influx of new accounts makes staffing optimization a challenge. Overstaffing leads to idle capacity and reduced ROI, while understaffing delays advertiser support and reduces incremental revenue.

Your goal is to develop a dynamic month-by-month staffing plan that ensures efficient support while optimizing costs.

2 Key Rules & Workflow

Advertiser eligibility

- When an advertiser signs up, Google projects its potential annual advertising budget.
- If the budget exceeds a country-specific threshold, the advertiser becomes eligible for agent support. Otherwise, it is ineligible and left untreated.

Agent assignment & capacity

- Eligible advertisers are placed in a waiting pool.
- If agent capacity is available, an agent is assigned.
- If no capacity is available, the advertiser waits up to 60 days in the pool.
- If still unassigned after 60 days, the advertiser is removed from the pool.

Agent lifecycle

- Each agent manages a maximum of 10 advertisers at any point.
- You can assume that all agents start 2025 with no advertisers already assigned to them.
- Advertisers receive support for 60 days before graduating, which frees agent capacity.
- New agents require 1 month of ramp-up time before they become available.
- Firing an agent requires 1 month of advance notice and costs 40% of the agent's annual salary.

Revenue impact

Agent support drives incremental advertising spend, distributed as follows:

Uplift in ad spend	Probability
0%	5%
5%	15%
10%	25%
15%	25%
20%	20%
25%	10%

3 The Challenge: A Month-by-Month Staffing Plan

Determine the optimal number of agents per country for each month of 2025, balancing:

- **Cost efficiency:** minimize idle agents while meeting advertiser demand.
- **Timely advertiser support:** reduce wait times and maximize revenue uplift.
- **Scalability:** adjust for market fluctuations given the hiring and firing constraints.

Recommended deliverables

- **Dynamic staffing model:** a structured plan showing how agent headcount changes month by month.
- **Analysis & justification:** a brief report explaining key decisions, assumptions, and the impact on revenue and costs.
- **Sensitivity analysis:** how the plan performs under different sign-up volume scenarios.
- **Other considerations:** any other factor you recommend considering when setting agent staffing levels.

4 Data Provided

The dataset is provided by the teaching team along with this brief. It contains:

Data	Details
Annual agent salary	Per country, in USD
Advertiser eligibility threshold	Per country, in USD
Advertiser sign-ups, 2023 and 2024	One row per sign-up: advertiser ID, country, date of sign-up, and projected annual advertising budget
Existing agent staffing levels	Number of agents per country as of January 2025

5 Making It a GenAI Application

The hackathon asks you to build a working AI application that combines **at least two course techniques** (API calls, prompt engineering, structured output, LangChain, RAG, or agents). The planning model above is the problem; your AI application is how you solve it and put it in people's hands. The directions below are examples, not requirements. Pick one, combine several, or propose your own.

AI ASSISTANT

Staffing Planner Assistant

A chat assistant that answers questions like "How many agents does each country need in March, and why?" It retrieves the case rules and your model's outputs, and calls your model as a tool to answer what-if questions.

Techniques: RAG, agents (tool calling), structured output

DATA ANALYZER

Sign-up Insights in Plain English

Ask questions about the sign-up data in plain English ("Which countries have the most volatile demand?") and get charts and short written insights back.

Techniques: Agents (code or tool use), prompt engineering, structured output

WORKFLOW AUTOMATOR

Scenario Explorer

A manager describes a scenario in words ("sign-ups in one country drop 30% in Q3"). An LLM turns it into structured model parameters, your model re-runs, and the app explains how the plan and costs change.

Techniques: Structured output, LangChain or agents, prompt engineering

CONTENT PIPELINE

Staffing Memo Generator

Turns each month's plan into a short memo per country manager: headcount changes, hiring and firing dates, expected costs and revenue uplift, and the reasons behind them.

Techniques: LangChain chains, prompt engineering, structured output

6 Scoping for One Week

- Start with one or two countries to get a simple month-by-month simulation working, then scale up to all of them.
- Work with expected values first (for example, the expected revenue uplift per supported advertiser) before adding randomness.
- Keep the GenAI component a working part of your product, connected to your model, rather than a separate add-on.
- The best hackathon projects are not the most complex. They are the most focused. A small model that works end to end beats a big one that doesn't run.

7 What to Submit

Due **Sunday, October 25, 11:59 PM ET** on Canvas (7 — Hackathon Project). This is a group assignment: one team member submits, and all members contribute to every deliverable.

- Your completed Jupyter notebook (`.ipynb`) with every cell run and all outputs visible.
- Your GitHub repo link, containing the code and a README with setup instructions, an architecture diagram, and each team member's contributions.
- A link to your 3-minute demo video walking through your product, your architecture, and the part you're most proud of.
- Review the rubric attached to the assignment to make sure your submission meets every requirement.

AI use. You may use AI tools, but explain how you used them and clearly mark AI-generated sections. You are responsible for making sure everything you submit is accurate and follows Northeastern's Academic Integrity and Use of AI policy (linked in the syllabus).

8 Choosing This Topic

Each team works on one topic. Topics are assigned **first come, first served** through the topic sign-up sheet, and each topic has a limited number of team slots. Check that this topic still has open slots, then register your team. Don't start building until your topic is confirmed on the sheet.

Questions about the topic? Post them in the Discussions Q&A Forum on Canvas so other teams can benefit, or bring them to office hours.